genstats / inst / doc / 04_10_eQTL.Rmd

jtleek 21 days ago updated eqtl/gene sets

1 contributor

193 lines (138 sloc) 4.6 KB

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<tr><td>eQTL</td><td>Jeff Leek</td><td><table><tr><th>rmarkdown::html_document</th><th>vignette</th></tr><tr><td><table><tr><th>toc</th></tr><tr><td>true</td></tr></table></td><td>%\VignetteIndexEntry{eQTL in R} %\VignetteEngine{knitr::rmarkdown} \usepackage[utf8]{inputenc}</td></tr></table></td></tr>
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Dependencies

This document depends on the following packages:

```
suppressPackageStartupMessages({
  library(devtools)
  library(Biobase)
  library(MatrixEQTL)
})
```

```
  library(devtools)
  library(Biobase)
  library(MatrixEQTL)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools","MatrixEQTL"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase"))
```

Download the data

Here we are going to follow along with the tutorial on MatrixEQTL. First we find the files

```
base.dir = find.package("MatrixEQTL")
SNP_file_name = paste(base.dir, "/data/SNP.txt", sep="");
expression_file_name = paste(base.dir, "/data/GE.txt", sep="")
covariates_file_name = paste(base.dir, "/data/Covariates.txt", sep="")
output_file_name = tempfile()
```

Next we load the data so we can see it

```
expr = read.table(expression_file_name,sep="\t",
                  header=T,row.names=1)
```

```
expr[1,]

snps = read.table(SNP_file_name,sep="\t",
                  header=T,row.names=1)
snps[1,]

cvrt = read.table(covariates_file_name,sep="\t",
                  header=T,row.names=1)
```

eQTL is linear regession

The simplest eQTL analysis just computes linear regression models for each SNP/gene pair.

```
e1 = as.numeric(expr[1,])
s1 = as.numeric(snps[1,])
lm1 = lm(e1 ~ s1)
tidy(lm1)
```

We can visualize the data and the model fits

```
plot(e1 ~ jitter(s1),
     col=(s1+1),xaxt="n",xlab="Genotype",ylab="Expression")
axis(1,at=c(0:2),labels=c("AA","Aa","aa"))
lines(lm1$fitted ~ s1,type="b",pch=15,col="darkgrey")
```

Fitting many eQTL models with MatrixEQTL

Set general parameters

We need to set up the p-value cutoff and the error model (in this case assuming independent errors)

```
pvOutputThreshold = 1e-2
errorCovariance = numeric()
useModel = modelLINEAR
```

Set the data up

Now we need to set up the snp and gene expression data in the special format required by the `MatrixEQTL` package

```
snps = SlicedData$new()
snps$fileDelimiter = "\t"     # the TAB character
snps$fileOmitCharacters = "NA" # denote missing values;
snps$fileSkipRows = 1          # one row of column labels
snps$fileSkipColumns = 1       # one column of row labels
snps$fileSliceSize = 2000     # read file in pieces of 2,000 rows
snps$LoadFile( SNP_file_name )


gene = SlicedData$new()
gene$fileDelimiter = "\t"      # the TAB character
gene$fileOmitCharacters = "NA" # denote missing values;
gene$fileSkipRows = 1          # one row of column labels
gene$fileSkipColumns = 1      # one column of row labels
gene$fileSliceSize = 2000      # read file in pieces of 2,000 rows
gene$LoadFile(expression_file_name)


cvrt = SlicedData$new()
```

Running MatrixEQTL

We can now run the code to calculate the eQTL that we are interested in

```
me = Matrix_eQTL_engine(
    snps = snps,
    gene = gene,
    cvrt = cvrt,
    output_file_name = NULL,
    pvOutputThreshold = pvOutputThreshold,
    useModel = useModel,
    errorCovariance = errorCovariance,
    verbose = TRUE,
    pvalue.hist = TRUE,
    min.pv.by.genesnp = FALSE,
    noFDRsaveMemory = FALSE);
```

Understanding the results

We can make a plot of all the p-values from the tests

```
plot(me)
```

We can also figure look at the number and type of eQTL

```
me$all$neqtls
me$all$eqtls
```

More information

eQTL is an entire field of research.

- MatrixEQTL package
- MatrixEQTL paper
- Surrogate variables applied to eQTL
- A Bayesian sva framework for eQTL
- Bioconductor eQTL workflow

Session information

Here is the session information

```
devtools::session_info()
```

It is also useful to compile the time the document was processed. This document was processed on: `r Sys.Date()` .